MACH1 / FleetLine E1 — Battery Section Executive Report

Prepared for: competition jury review **Date:** 16 October 2026 **Status:** Final conclusions from full literature audit, primary-source market verification, and an independently audited lifecycle model **Companion:** `MACH1_BATTERY_SLIDES_CONTENT.md`

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0. The one-paragraph version

We chose **LFP over sodium-ion** because three independent walls — no 45 kWh-class supplier, no cycle data at Indian ambient, and Indian homologation standards that name *lithium* in the title — each independently block certification before the 2034–35 launch window. None of those walls is about cost. We chose **LFP with a detachable pack architecture** so the chemistry stays swappable and sodium remains a genuine 2036–38 option rather than a written-off bet. On sizing, the honest answer is a **range, not a number: 40.6–74.6 kWh** depending on two things published LFP data cannot resolve. The decisive finding of this work is that **cell temperature, not pack size, determines whether the 8-year warranty holds** — and we can demonstrate it with arithmetic on peer-reviewed LFP data rather than assertion. That reframes the deliverable from *"a pack"* to *"a thermal architecture plus a costed validation programme."*

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1. Chemistry selection

1.1 The original proposal's thesis, stated fairly

The MACH1 deck (`MACH1_NITDURGAPUR.pdf`) proposes:

> *"architecture designed around Na-Ion Battery. The car targets 2035, We have enough runway for Na-Ion to close the cost gap and mature"*
> *"Na-Ion commercially finalised around Na-Ion primary cell" … "Na-Ion as primary offering, LFP as [fallback]"*

The survey document already hedges: *"We will consider battery pack, LFP type. Predict Na-ion to reach LFP price range in 5 yrs after"*.

This is not a naive claim, and it deserves a fair hearing. Sodium-ion's cost trajectory is real, and layered-oxide sodium cells genuinely do reach LFP price parity on a five-year view. The instinct — *"we launch in 2035, so wait for the better chemistry"* — is commercially reasonable.

It fails on **timing and certification**, not on direction.

1.2 Why 2035 does not solve it — three independent walls

Each of these must clear. Any one of them failing kills the primary-chemistry plan.

Wall 1 — The cell does not exist at our scale. No supplier ships layered-oxide sodium-ion in a sedan-class vehicle. Current automotive deployments are **A00-segment microcars at 23–25 kWh** (HiNa in the JAC Yiwei 3, Farasis in the JMEV EV3). The one sodium-ion product with GWh capacity *and* a passenger car *and* GB 38031-2025 certification is CATL Naxtra — and it is **Prussian White**, a different cathode family from the layered oxide the proposal specifies. Prussian White is not a drop-in substitute for the specified chemistry.

Wall 2 — The data does not exist. There is **zero cycle-life data for any layered-oxide sodium cell at 60–80% DoD *and* 40–45 °C.** Every public headline claim has undisclosed DoD, undisclosed temperature, or both. RWTH tested commercial layered-oxide cells to 45 °C but only for *calendar* ageing; TUM found severe 45 °C degradation at high SOC, again without cycling. By contrast LFP has published, citable data: **4,000 cycles at 25 °C, 2,000 at 45 °C** (WRI). We cannot build a defensible comparison when one side has no number at all.

Wall 3 — It cannot be legally homologated in India. AIS-038 Rev.2 Amendment 3, Annex IX-K(2), verbatim:

> *"Cells used to make REESS, shall be certified as per IS 16893-Part 2 and Part 3 by NABL accredited lab or by test agency notified under CMV Rule 126."*

IS 16893-2 and -3 are titled *"Secondary **Lithium-Ion** Cells…"*. No amendment through Amd 4 has added an alternative-chemistry clause or scope exemption. No sodium-ion cell has ever been certified to them.

There is no import workaround. Importing a finished pack does not help, because the pack itself must be type-approved under AIS-038, which in turn requires the cell certification.

A year of additional runway does not move Wall 3. That clause names lithium in the standard's own title. It is a standards question, not a maturity question. Waiting does not fix it — only a standards amendment does.

Joint probability of all three walls clearing by January 2034: under 10%.

1.3 The duty and tariff detail that cuts *against* sodium

Item	Position	Effect on sodium
HSN 8507.60	*"Lithium-ion accumulators"* only	Na-ion falls under **8507.80 "Other accumulators" → 15% BCD** vs 20%
5% concessional BCD	Operative text: *"for **Lithium-Ion** Cells for batteries of EVs or BESS"*	**Excludes sodium**
0% capital-goods BCD	*"for **Lithium Ion Cell** manufacturing"*	**Excludes sodium**
GST	Uniform **18%** across heading 8507 post-GST-2.0; PIB explicitly names sodium in the 28%→18% cut	Neutral

Net effect: sodium gets a marginally lower headline duty (15% vs 20%) but **loses every concession**, and GST is neutral.** There is no duty advantage either way.** The lower headline number is misleading unless stated alongside the lost concessions.

1.4 Decision

LFP is the primary chemistry. Sodium-ion is a documented 2036–38 second act — retained precisely because the architecture keeps it available. We are not arguing sodium-ion is the wrong direction. We are arguing it cannot be *certified* on the proposed date, and a programme that cannot be certified cannot be delivered.

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2. Architecture — why detachable, not structural

Semi-structural CTP · standard prismatic envelope · chemistry-specific firmware · common BMS hardware.

Dimension	Detached semi-structural CTP	Full structural CTC
Chemistry change	Cell swap + firmware	Effectively impossible
AIS-038 Clause 2.48 retest	4–6 months, ₹2–3 Cr	18–24 months, ₹15–25 Cr
Cell standardisation	256×106×32 mm prismatic, 102–108 s	Chemistry-locked
Repurposing / second life	Straightforward	Very difficult
Body programme impact	None	Full body-in-white retool

The argument in one line: Clause 2.48 requires a full REESS retest on *any* chemistry change. That cost is unavoidable. The architectural question is only whether the *cell* stays swappable afterwards. Detachable says yes; structural says never.
This also creates a **residual-value argument** for judges: a buyer can upgrade cells without buying a new car.

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3. Market analysis

3.1 Two claims we removed, and why

Removed claim	Finding
Xpres-T fleet share **89% → 53%**	Tata press releases return 403. ET, Autocar Pro and Financial Express all redirect to an unrelated article. **No Vahan or SIAM corroboration exists.** Tata's FY24 annual report does not disclose fleet-EV share. The definition is ambiguous — 89% of *fleet EV registrations*, *all EV passenger cars*, or *Tata's own fleet EV sales*? Unknowable
Fleet orders **26,000 → 2,000** (−92%)	The cited Financial Express article **does not exist.** Every URL ending `3400000` redirects to a PM Modi tribute article. Wayback Machine has no capture. **Zero corroboration** from SIAM, Vahan, Redseer, ICCT, Jefferies or Crisil

This is not fixable with more digging. Indian fleet-EV share is not reliably measurable from public data: Vahan aggregates by category (M1, taxi, maxi-cab) and never provides an OEM- or model-level fleet/private split; fleet-versus-private classification is inconsistent across states; retro-fit and grey-market registrations go uncounted. JMK, IEA and SIAM all derive from the **same Vahan feed** — they are one source dressed as three.

3.2 What we can defend instead

TCO parity already exists. Built entirely from primary sources — OEM fleet price lists (2025), state electricity tariffs (DERC/BESCOM/TNERC/MERC), CNG prices (IGL/Mahanagar Gas/ APCPDCL), 9.5% fleet finance rate, 4-year / 1.2 lakh km, 300 km/day utilisation with depot charging:

City	45 kWh EV sedan	CNG sedan	Advantage
Delhi	**₹2.55/km**	₹3.15/km	−19%
Bengaluru	**₹2.80/km**	₹3.60/km	−22%
Mumbai	**₹2.95/km**	₹3.80/km	−22%
Hyderabad	**₹2.65/km**	₹3.35/km	−21%

Published failure mode: below ~150 km/day utilisation, or without depot charging, CNG wins. We state this alongside the table — a TCO claim without its boundary condition is not a claim.
The 2025–26 dip has a documented cause that needs no fabricated statistic.

Scheme	4W passenger coverage	Fleet eligible
FAME II (ended 31 Mar 2024)	☐ ₹10,000/kWh, capped at **20% of vehicle cost**; vehicle price must be <₹15 lakh	☐
PM E-DRIVE (from 29 Sep 2024, **extended to 31 Mar 2028**)	☐ **4W passenger cars excluded**	☐
PM E-DRIVE scope	e-2W, e-3W (e-rickshaws/e-carts, L5), e-ambulance, e-truck only	—

*Correction: FAME II's ₹1.5 lakh figure was **not** the 4W cap — that is the e-2W *price threshold*. The 4W cap is 20% of ex-factory cost, and NAB records actual e-Verito payouts of ₹1.59–2.12 lakh.*
The fleet-EV slowdown tracks a subsidy cliff and an explicit policy exclusion. That is a better explanation than a 92% collapse, and it is verifiable from the gazette.

Mandates are real and gazetted. The Delhi Motor Vehicle Aggregator and Delivery Service Provider Scheme, 2023 (gazette SG.DEL-E.29112023-250308) sets **new-fleet onboarding rates**, not a total-fleet conversion deadline:

Four-wheeler	6 mo	1 yr	2 yr	3 yr	4 yr	5 yr
EV share of new fleet onboarded	5%	15%	25%	50%	75%	**100%**

Separately, every aggregator and delivery fleet must be **fully electric by 1 April 2030**. The scheme **does carry penalties** — Chapter V permits licence suspension or cancellation for non-compliance — so this is an enforced ladder, not an aspiration.
Correction: an earlier draft of this document cited these as "5% by 2025, 25% by 2027, 100% by 2030". Those percentages were right but attached to the wrong years; the schedule above is the gazette's actual milestone structure.

The real buyer base is small, verifiable, and multi-sourced.

Operator	Total fleet	EV fleet	EV %	Source
Everest Fleet	**22,000+**	~2,000	~9%	ICRA rating rationale, 31 Dec 2025
Lithium Urban	3,000+	3,000+	~100%	Company disclosure
BluSmart	—	—	—	**Defunct** — operations collapsed 2025
Mahindra Logistics (Meru→Alyte)	asset-light, not disclosed	not disclosed	—	Asset-light; fleet is largely driver-owned
Ola / Uber	platforms, not owners	Not disclosed	—	—

*Corrections: Everest's own website still says 18,500 vehicles / 2,000+ EV; the ICRA rating rationale gives **22,000+ as of 31 Dec 2025**, which we use as the current figure. BluSmart is not merely "paused"

— it is **defunct**. The previously cited "50,000+ deployed / ~500 EV" for Mahindra Logistics could not be sourced and has been removed; the company runs an asset-light model and does not publish EV counts.* Addressable owned/managed fleet base is therefore **order-of-magnitude 40,000–50,000 vehicles, with roughly 8,000–10,000 electric**. Growth since 2023 has come from dedicated EV fleetcos rather than aggregator conversion — the direction that matters for us.

Multi-sourcing is the norm. Everest runs Tigor EV, Xpres-T and Nexon EV simultaneously. Meru specifies e-Verito, Tigor, Nexon and MG ZS. State tenders (KSRTC, MTC, DTC) specify technical parameters, not OEM. **There are no platform lock-ins.** Whoever delivers the best TCO wins the next tender.

Competitive outlook. Maruti Suzuki has no fleet-EV programme. The eVX is a **60 kWh private SUV positioned as a global export flagship** (Europe and Japan), with published price estimates spanning **₹18–25 lakh** depending on source and variant. That sits decisively above the ₹10–14 lakh fleet-sedan band we intend to compete in.

*Correction: an earlier draft claimed the eVX was "unlikely to contest the segment before 2028". Published launch estimates conflict widely — from late 2026 to early 2029 — so the timing is not something we can assert. The **price band** is the defensible point: even at its cheapest estimate, the eVX is ~30% above our band.*

2034 addressable volume: 60,000–80,000 units/year for a competitive sedan. *(All volume forecasts are secondary/analyst — no official government forecast exists. Labelled as such.)*

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4. Duty cycle — verified, not assumed

Quantity	Value	Source
Base consumption	12.84 kWh/100 km	Corrected from deck's 15.38
Daily energy	**25.68 kWh/day**	Real-world verification
Annual equivalent full charges	**~200 EFC/yr** (190–228 range)	Corrected from deck's 365
Deck's original figure	38.28 kWh/day	**~49% overstated** vs verified

The deck originally claimed HVAC alone was "2.24× overstated." That framing is wrong — the overstatement is ~49% on total daily energy. Corrected.

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5. The lifecycle model

5.1 Structure

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SoH(t, EFC) = 1.02 − A_cal·t^α·Arrhenius(T)·k_SOC − k_cyc·DoD^β·Arrhenius(T)·(EFC·8)
```

Term	Meaning	Primary source
`A_cal`	calendar capacity-fade coefficient	**Naumann 2018** (`10.1016/j.est.2018.01.019`)
`α`	calendar **time exponent — contested, 0.45–1.00**	Lewerenz 2017 vs Gasper 2021
`β`	**DoD exponent — contested, −0.40 to +1.00**	Schreiber / Gauthier / Eldesoky
`Ea`	activation energy	Naumann: 17.126 kJ/mol (capacity), 71.827 kJ/mol (resistance)
`k_SOC`	SOC dependence	Naumann cubic, `0.6140` at 66% mean SOC (literal)

5.2 Calibration provenance — Naumann 2018

Cell: Sony/Murata **US26650FTC1**, 3 Ah, 26650 cylindrical **LFP/graphite** — the same cell family Graner used. 885 days, 17 test points × 3 cells, <2.2% max error on capacity fade.

A normalisation detail worth stating, because it was one of our own errors and it changes the numbers: Naumann's Eq. (6) divides `k_SOC(SOC_ref)` **out** of `k_ref`, so the forward model multiplies by `k_SOC(SOC) = 0.6140` at 66% mean SOC. An earlier draft used the renormalised ratio `0.6399` instead — the two are **not** equivalent (4.06% on calendar fade, **28.72% on resistance growth**).

5.3 Default results (Naumann literal, DoD^1.0, α = 0.50, linear superposition)

Cell temp	Pack	EFC/yr	EFC @ yr 8	SoH @ yr 8	R_inc @ yr 8
48 °C	**44.2 kWh**	193	1548	71.6%	**+101.3%**
40 °C	**41.3 kWh**	207	1657	76.7%	+50.9%

5.4 Full parameter sweep — 48 °C cell

DoD exp	α = 0.45	α = 0.50	α = 0.75	α = 1.00
−0.40	50.0 kWh / 63.5%	51.3 / 61.8%	62.9 / 50.4%	**100.1 / 31.7%**
0.00	47.1 / 67.3%	48.3 / 65.7%	58.0 / 54.7%	87.6 / 36.2%
+0.25	45.9 / 69.1%	**46.9 / 67.6%**	55.9 / 56.7%	83.2 / 38.1%
+0.50	44.8 / 70.7%	45.9 / 69.1%	54.4 / 58.3%	80.1 / 39.6%
+1.00	43.3 / 73.2%	**44.2 / 71.6%**	52.1 / 60.8%	76.1 / 41.7%

Bold = model default. Full 40 °C sweep also computed.

5.5 Thermal scenario envelopes

Scenario	Cell	Pack range	SoH range	R_inc @ yr 8
Immersion *(indicative)*	26 °C	37.1–53.3 kWh	59.5–85.5%	+14.0%
Proven chiller loop	**40 °C**	**40.6–74.6 kWh**	**42.5–78.1%**	**+50.9%**
Liquid plate	**48 °C**	**43.3–100.1 kWh**	**31.7–73.2%**	**+101.3%**
Passive / air only	55 °C	46.2–**140+** *(no convergence)*	22.0–68.6%	+179.8%

5.6 The 55 °C result is a specification failure, not a sizing problem

At 55 °C cell the DoD exponent is **−0.40**, so *shallow* cycling becomes the **worst** case — the exact opposite of 40 °C. Combined with the linear calendar exponent, the worst corner exceeds the model's 140 kWh search ceiling.

At 55 °C cell the duty is not achievable at any pack size the model can search. We deliberately did **not** raise the ceiling to manufacture a finite number. The infeasibility is the finding. The deck's 45 kWh sedan is out by roughly **3×**.

5.7 An independent cross-check on the cell the model is calibrated to

Our entire calibration rests on one cell: the **Sony US26650FTC1** LFP/graphite cell in Naumann 2018. That is a real exposure, so it is worth showing that cell has been independently benchmarked by an unrelated group.

Harlow et al. 2019 (*J. Electrochem. Soc.* 166(13) A3031–A3044, `10.1149/2.0981913jes`) took Dalhousie's state-of-the-art NMC532/graphite pouch cell and tested it for up to three years at **20, 40 and 55 °C**, both cycling and storage — and plotted its own storage fade **directly against Naumann's LFP/graphite Sony US26650FTC1 cells** in Fig. 3, the same cell our model uses.

Two things follow, and they point in opposite directions:

- **It validates the reference cell.** Our cell is the field's benchmark reference, not an obscure dataset. Harlow treats it as the comparison baseline precisely because it is well-characterised.
- **It does not transfer to us.** Harlow's 1.6 million km headline belongs to an NMC cell cycled to a **4.3 V** upper cutoff — a higher ceiling, a better-sealed pouch format, and a chemistry with a different degradation profile. It is an **upper bound for the best benchmark cell in the field**, not a number our LFP pack can claim.

Harlow also states plainly that *"the cycle and calendar lifetime of LFP/graphite cells are superior to that of NMC/graphite cells"* — a same-field bridge from the reference cell to our chemistry. **It still contains no LFP depth-of-discharge matrix**, so §6.2 remains open.

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6. The two contested parameters

6.1 Calendar time exponent — 2.3× spread

Source	Chemistry	Method	α
Liu 2018	LFP	fitted power law	0.4155
Gasper 2021	NMC + LFP pooled	statistical validation, 17 datasets	~0.45
Naumann 2018	**LFP**	fitted √t	0.50
Lewerenz 2017	**LFP**	**measured linear fade, 50 cells**	**1.00**

Gasper's finding is that a **fixed** t^0.5 is statistically invalid — only **1 of 17** datasets contained 0.5 within its own 95% confidence interval, despite $R^2 \geq 0.99$. But Gasper also finds the exponent **rises toward 0.5 as temperature increases**, which pulls against Lewerenz at our operating point. The literature does not converge.

6.2 DoD exponent — small, direction-ambiguous, and not a temperature function

Source	Chemistry	Cell temp	UCV	β
Schreiber 2025	NMC/graphite	20 °C	—	0.00 (flat)
Gauthier 2022	NMC622/NG	20 °C	4.10 V	non-linear
Gauthier 2022	NMC622/NG	**40 °C**	4.10 V	**+1.00**
Eldesoky 2022	**NMC811/AG**	**20 & 40 °C**	**4.06 V**	**no dependence at all**
Eldesoky 2022	**NMC811/AG**	**20 & 40 °C**	**4.20 V**	**small, slightly INVERSE**
Eldesoky 2020	**LFP**	55 °C	3.65 V	**−0.40**
Our interpolation	—	*48 °C*	—	*+0.25 — ASSUMPTION*

This kills the temperature hypothesis. We had modelled the DoD exponent as crossing zero with temperature. Eldesoky 2022 shows two NMC papers at **the same 40 °C** giving opposite answers: Gauthier linear **+1.0**, Eldesoky small and **inverted**. The literature disagrees even at fixed temperature. Our interpolated zero-crossing near 50.7 °C is an artefact of stitching different chemistries at different voltages, not a physical law.

Eldesoky 2022's headline result is that the DoD effect is simply SMALL:

> *"Overall, the impact of DOD on capacity loss and ΔV growth was small… **differences between the various DODs in Fig. 4 were within 10–15 mAh which is small**."* — across the entire 25→100% DoD range, after ***12,000 hours*** of cycling.

6.2.1 The reconciling variable is upper cut-off voltage, not temperature

Eldesoky 2022 varied the **lower cut-off** to set DoD (25/50/75/100%) against two upper cut-offs, and the UCV determined whether a DoD effect existed at all:

UCV	DoD / C-rate dependence
4.06 V	**None.** *"the 4.06 V cells showed no C-rate- or DOD-dependency"* for impedance growth
4.20 V	Small and **inverse**: *"larger DODs generally resulted in smaller capacity loss"*

The stated mechanism is identical to Eldesoky 2020's:

> *"a slight increase in capacity loss with ***smaller*** DODs which we attributed to the fact that the cells spent ***more time at higher voltages*** when the DODs were small"*

So DoD is a weak lever; high-voltage dwell is the strong one. Two independent studies, different chemistries, different voltage windows, reach the same mechanism.

6.2.2 Why this is a strong argument *for* LFP

LFP's charge ceiling is ~3.6 V — **below Eldesoky's 4.06 V threshold at which DoD and C-rate dependence vanish entirely.** We are not choosing to limit the DoD dependence; our chemistry is already below the level where it appears.

It also validates the design requirements already in the document (`CW-1…CW-5`): a ceiling on time-at-high-SoC, a charge window, and mandatory relaxation. Those are not housekeeping — they are the levers that actually control degradation in this evidence.

Post-mortem confirmation: *"no separator discoloration or signs of Li plating on the anode"* (2% VC + 1% DTD electrolyte). The cutoff-based DoD variation was **not** confounded by plating — the confound we were worried about is ruled out.

6.3 What this means for the model

β is secondary. The calendar exponent α dominates. Holding α = 0.50 at 48 °C:

β	Pack
−0.40	51.3 kWh
0.00	**48.3 kWh**
+1.00	44.2 kWh

A 7 kWh spread across the *entire* β range. Holding β = +1.00 and varying α: **44.2 → 76.1 kWh**, a 32 kWh spread. Eldesoky 2022 independently supports β near 0.
Revised position: β should be treated as a small parameter near zero, not a free variable swept across ±1. The α dispute remains the dominant open question, and Eldesoky 2022 does nothing to resolve it.

6.4 What this means for the proposal

We cannot resolve either parameter from published LFP data at our operating point. We could pick a value, but picking one and presenting it as fact would be the specific failure mode this audit was commissioned to eliminate.

The engineering response is the validation programme in §8, not a better guess.

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7. Two findings that changed the proposal

7.1 The literature disagreement, and why it is decisive

Two peer-reviewed studies. Same LFP chemistry. Same 40 °C. Opposite answers:

Study	Method	Time dependence	Years to 80% SoH at 40 °C
Naumann 2018	Fitted √t law	t^0.5	**13.3 yr**
Lewerenz 2017	**Measured linear fade rate**	**t^1.0**	**3.7 yr**

Lewerenz measured linear fade at 40 °C of **1.50–1.90% per 100 days** across 20–100% SOC, with weak SOC dependence (~7% from 50% to 100% SOC) — which independently corroborates Naumann's flat-mid-range finding. Their own extrapolation: **">3–4 years storing at 40 °C."**
Our model reproduces 3.69 years from Lewerenz's own Table 6 slopes. That is a cross-check, not an assertion — and it reproduces the *conservative* paper, not the favourable one.

The paradox that makes this obvious in hindsight: Lewerenz is **slower in year one** (5.96%/yr vs Naumann's 9.8%/yr) and **far worse by year eight**. Long-horizon life is governed entirely by the exponent, not the year-one rate.

> ***Anyone who validates at year one and extrapolates linearly will get the 8-year answer backwards.** This is the single most important thing we learned.*

Consequence: the previously derived "≤40 °C cell is a hard requirement" is **not sufficient**. Under the Lewerenz exponent, an 8-year / 80% warranty is unreachable at 40 °C on calendar ageing alone. Either the warranty target drops, the temperature envelope tightens well below 40 °C, or the claim ships with the spread attached.

7.2 Resistance growth — an exposure nobody had modelled

Naumann's measured activation energy for **resistance** growth is **71.827 kJ/mol — 4.2× the 17.126 kJ/mol for capacity.**

Cell temp	R_inc @ yr 8	Implied peak power
40 °C	+50.9%	~66% of new
48 °C	**+101.3%**	**~50% of new**

At 48 °C the pack reaches **80% SoH on paper while delivering roughly half its rated power.** Naumann's resistance term is **linear in time**, not √t — so it does not share the calendar exponent controversy, and this exposure is present under *every* parameter combination in the sweep.
A capacity warranty covers none of this. An 8-year fleet vehicle delivering half its rated power is a warranty event that state-of-health cannot see. **This requires its own warranty clause** — a power-delivery threshold, not a capacity threshold.

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8. Validation programme

Six cells · 40–45 °C · 90–100% SoC · ≥18 months · replicated.

8.1 Why these exact parameters

Parameter	Value	Justification
Cells	6 (from 3)	Spread uncertainty scales as 1/√(2(n−1)): **~50% at n=3 → ~32% at n=6**
Temperature	40–45 °C	Our realistic operating point, and the band where literature disagrees most
SoC	90–100%	Where calendar fade is strongest; Graner's lab matrix had no points above 87.5% at 40 °C
Duration	≥18 months	Must span at least one seasonal cycle

8.2 The quantitative justification

Graner et al. (arXiv 2609.10185) built a probabilistic LFP degradation framework on 17 calendar + 19 cyclic conditions, 3 replicates each, 885 days, validated against 21 residential systems over 5.2 years.

Their full predicted end-of-life range was ~3 years on a 12.3-year mean — about 33% of lifetime. And critically:

> ***One outlier cell at a single frequently-visited condition (25 °C / 100% SoC, 5.9% spread) accounted for ~50% of the entire EOL range.** Replacing it narrowed the range from 3 years to 1.5 — a **50% reduction**.*

This is the argument judges cannot dismiss: our gap is not rhetorical. In a better-instrumented dataset than ours, a single anomalous cell moved the answer by half. Without replication we cannot even *detect* such a cell.

8.3 What this one programme closes

Gap	Closed by
Calendar exponent (0.45 vs 1.0)	High-SoC holds at our temperature, past 18 months
Activation energy	40–45 °C cell data at our SoC
SOC dependence	90–100% SoC arms
Replicate-derived outlier risk	n = 6 vs n = 3

This converts our largest weakness into a costed, dated, defensible plan. That is the single strongest element in the proposal.

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9. What we will not claim

Claim	Status	Reason
"8-year / 80% SoH"	**Not defensible at 40 °C**	Lewerenz vs Naumann disagreement unresolved
Xpres-T fleet share 89%→53%	**Removed**	No primary corroboration exists
Fleet orders −92%	**Removed**	Cited article does not exist
A single pack size	**Removed**	Range is the honest answer
Market-share argument	**Replaced**	TCO parity + gazetted mandates, both primary
Na-ion as primary chemistry	**Replaced with LFP**	Three independent certification walls
Charge rate as a design constraint	**Removed**	Not a real constraint at our duty
"HVAC 2.24× overstated"	**Corrected**	~49% on total daily energy, not HVAC alone

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10. Appendices

Appendix A — Evidence base (all DOIs verified against Crossref)

Source	Citation	Role
Naumann et al. 2018	*J. Energy Storage* 17, 153–169 · `10.1016/j.est.2018.01.019`	LFP calendar model: Ea, SOC factor
Lewerenz et al. 2017	*J. Power Sources* · 50-cell LFP dataset	Linear fade measurement at 25/40/60 °C
Simolka et al. 2020	*J. Appl. Electrochemistry* 50, 1101–1117 · `10.1007/s10800-020-01465-6`	LFP DoD + temperature post-mortem
Schreiber et al. 2025	*J. Energy Storage* 112, 115357 · `10.1016/j.est.2025.115357`	Realistic VW ID.3 duty cycling
Gauthier et al. 2022	*J. Electrochem. Soc.* 169, 020518 · `10.1149/1945-7111/ac4b82`	DoD × C-rate × calendar matrix
Gasper et al. 2021	*J. Electrochem. Soc.* 168, 020502 · `10.1149/1945-7111/abdde1`	t^0.5 statistical invalidity
Eldesoky et al. 2022	*J. Electrochem. Soc.* 169, 100531 · `10.1149/1945-7111/ac99a6`	DoD × C-rate × voltage × temperature
Graner et al. 2026	arXiv `2609.10185`	Lab-to-field uncertainty, probability bands
Waldmann et al. 2014	*J. Power Sources* 262, 129–135	25 °C temperature mechanism crossover
Pozzato et al. 2023	*Joule* · `10.1016/j.joule.2023.07.018`	Real-world field behaviour
Harlow et al. 2019	*J. Electrochem. Soc.* 166(13) A3031–A3044 · `10.1149/2.0981913jes`	3-yr NMC532 benchmark; plots itself against Naumann's LFP reference cell

Outstanding gap: Louli & Dahn / Harlow et al. 2019 LFP DoD matrix (`10.1149/2.0981913jes`) — the supplied file is Harlow et al. 2019, **NMC532**, not LFP.

Appendix B — Corrections made to the original proposal

#	Original	Correction
1	38.28 kWh/day consumption	**25.68 kWh/day** real-world (−33%)
2	"HVAC 2.24× overstated"	~49% on **total** daily energy
3	365 EFC/year	**190–228 EFC/year**
4	45 °C ageing claim	Restated on published Arrhenius basis
5	Charge-rate constraint row	**Removed**
6	Reliance / Faradion milestone	**Replaced**
7	Single 40 kWh pack size	**Range** across measured parameter space
8	8-year / 80% SoH warranty	**Qualified** — not defensible at 40 °C without validation
9	Xpres-T fleet share 89%→53%	**Removed** — no primary corroboration
10	Fleet orders −92%	**Removed** — source article does not exist
11	Na-ion as primary chemistry	**LFP primary**, sodium as documented 2036–38 option
12	—	**Resistance-growth warranty clause added** (never previously modelled)

Appendix C — Model provenance and error log

All figures reproduce from `/tmp/opencode/mach1/mach1_model.py`, independently verified by `/tmp/opencode/mach1/self_audit.py` **(exit 0)**.

- Default calibration: **Naumann 2018, literal SOC convention** (verified against Eq. 3 + 6)
- Calendar exponent sweep: **0.45 / 0.50 / 0.75 / 1.00**
- DoD exponent sweep: **−0.40 / 0.00 / +0.25 / +0.50 / +1.00**
- Hard model floor: **25 °C cell** (Waldmann mechanism crossover; `solve_min_pack` raises below)
- Structural uncertainty band: **40–50% on the 8-year EOL claim** — Graner-derived lower bound for data this sparse, reported separately and never merged into the parameter sweeps

Two errors of ours are preserved in the model's correction log rather than silently overwritten, so the chain stays auditable:

1. **Naumann SOC normalisation** — we used the ratio (0.6399) instead of the paper's literal value (0.6140). Error size: **4.06% on calendar fade, 28.72% on resistance growth.** The discarded form also manufactured a false finding (66% SoC appearing 2.07× worse than 100% for resistance). The audit carries a standing correction so it cannot be re-quoted.
2. **Two wrong DOIs.** Gasper 2021 and Naumann 2018 citations carried incorrect DOIs in

earlier drafts. In both cases the **article numbers were correct** and only the DOIs were faulty. Naumann's erroneous DOI resolved to an unrelated pumped-storage paper. Both now verified; the dead DOIs remain in the log for traceability.

Method note. Where a specialist agent's conclusion was checked and found incorrect, the correction was made against the primary source and the disagreement recorded rather than quietly overwritten — twice in this project, and both corrections are in the artifacts.